

Community Hero – AI-Powered Hyperlocal Problem Solver

Problem Statement Selected

Community Hero – Hyperlocal Problem Solver

Communities frequently face civic issues such as potholes, water leakages, fallen trees, damaged streetlights, waste accumulation, flooding, electrical hazards, and other public infrastructure challenges. Existing reporting systems are often fragmented, lack transparency, and provide limited visibility into issue resolution.

Community Hero addresses this challenge by enabling citizens to intelligently report, verify, track, and monitor community issues through AI-powered automation, real-time location awareness, and collaborative participation.

Solution Overview

Community Hero is a full-stack AI-powered civic engagement platform designed to modernize how local infrastructure issues are reported and resolved.

Citizens can instantly report civic hazards using live GPS location, photographs, and natural language descriptions. Google Gemini analyzes both images and text to automatically classify the issue, generate structured incident summaries, assess severity, recommend safety precautions, and provide engineering-focused municipal recommendations.

The platform supports community verification, real-time issue tracking, municipal workflow simulation, predictive infrastructure insights, and citizen engagement through gamification.

To ensure reliability, the system incorporates an intelligent server-side fallback engine with category-aware analysis templates, allowing the platform to continue delivering meaningful AI-assisted results even during temporary AI service interruptions.

The solution demonstrates how Generative AI can improve civic reporting accuracy, accelerate municipal response, and promote transparent collaboration between citizens and local authorities.

Key Features

Live GPS Location Detection

- High-accuracy browser GPS integration
 - Automatic reverse geocoding using OpenStreetMap Nominatim
 - Real-time location identification
 - Dynamic map centered on the user's live position
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AI-Powered Multimodal Issue Reporting

- Upload issue photographs
 - Natural language incident descriptions
 - Google Gemini Vision analyzes uploaded images
 - Automatic issue categorization into multiple civic issue types
 - AI-generated professional incident titles
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Intelligent AI Analysis

For every reported issue, Gemini automatically generates:

- AI Summary
- Issue Classification
- Confidence Score
- Severity Assessment
- Resolution Priority
- Actionable Safety Recommendations
- Vision Verification Checklist
- Estimated Dimensions (where applicable)
- Potential Risk Analysis
- Suggested Municipal Action
- Technical Engineering Analysis

Supported issue categories include:

- Roads & Potholes
- Fallen Tree / Road Obstruction
- Water Leakage
- Streetlight Failure
- Waste Management
- Flooding / Waterlogging
- Open Manhole

- Electrical Hazard
 - Generic Infrastructure Issues
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Community Verification

- Citizen upvote verification
 - Community-based issue validation
 - Automatic transition from Reported → Verified
 - Increased reporting transparency
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Municipal Workflow Management

Complete issue lifecycle management:

- Reported
- Verified
- In Progress
- Resolved

Includes a Municipal Control Panel Simulator for demonstrating real-world government workflows.

Predictive Infrastructure Insights

Gemini analyzes community issue trends to identify:

- Recurring road failures
- Water pipeline risks
- Flood-prone regions
- Infrastructure deterioration
- Public safety hotspots

This enables proactive maintenance instead of reactive repairs.

Citizen Engagement & Gamification

- Experience Points (XP)
- Community badges
- Citizen contribution tracking

- Leaderboard-ready engagement model
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Intelligent Reliability & Self-Healing Backend

To ensure uninterrupted service:

- Server-side Gemini response caching
 - Intelligent API quota management
 - Self-healing fallback engine
 - Category-aware AI enrichment
 - Weighted fallback classification
 - Robust handling of API quota exhaustion (429)
 - Graceful recovery from temporary service unavailability (503)
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Technologies Used

Frontend

- React
- TypeScript
- Vite
- HTML5
- CSS3
- SVG Interactive Maps

Backend

- Node.js
- Express.js
- TypeScript

Artificial Intelligence

- Google Gemini 2.5 Flash
- Gemini Vision (Multimodal AI)
- Server-side AI analysis engine
- Category-aware template registry
- Weighted fallback classifier

Location Services

- Browser Geolocation API
- OpenStreetMap Nominatim Reverse Geocoding API

Development Tools

- Visual Studio Code
 - npm
 - Git
 - GitHub
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Google Technologies Utilized

The project is built around Google's AI ecosystem and cloud infrastructure.

Google AI Studio

- AI application development
- Prompt engineering
- Rapid prototyping
- Testing and deployment

Gemini 2.5 Flash

- Intelligent issue categorization
- Natural language understanding
- AI-generated summaries
- Safety recommendations
- Technical infrastructure analysis
- Resolution prioritization

Gemini Vision

- Multimodal image understanding
- Infrastructure damage recognition
- Visual issue verification
- Context-aware hazard analysis

Google GenAI SDK (@google/genai)

- Secure server-side Gemini API integration
- Structured AI content generation

- Multimodal request handling

Google Cloud

- Application deployment using Google AI Studio Publish
 - Public cloud-hosted production deployment
 - Secure and scalable hosting environment
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Project Impact

Community Hero demonstrates how Generative AI can transform civic issue management by combining multimodal AI, live geolocation, intelligent automation, and community collaboration into a single unified platform.

The solution enables citizens to submit higher-quality reports, helps municipalities prioritize critical infrastructure issues, improves transparency throughout the issue lifecycle, and supports proactive infrastructure maintenance through AI-driven insights.

By integrating Google's latest AI technologies with real-world civic workflows, Community Hero delivers a scalable and intelligent approach to building smarter, safer, and more connected communities.